

This note contains a short analysis of certain expected quantities, as they appear in so-called randomized incremental construction. We adopt Mulmuley's terminology of *configuration spaces*, which we briefly review: S is some finite ground set containing n elements; C is some finite set of "configurations" with two functions $st, tr : C \mapsto 2^S$, with $st(c) \cap tr(c) = \emptyset$ for each $c \in C$. The elements in $tr(c)$ are the triggers of configuration c , and the elements in $st(c)$ are its stoppers. We will use $\tau(c) = |tr(c)|$ and $\sigma(c) = |st(c)|$, and we assume that for all c we have $\delta \leq \tau(c) \leq d$ for some constants δ and d . Configuration c is called *active* for $R \subset S$ iff R contains all of c 's triggers but none of its stoppers; c becomes active for some enumeration of S if c is active for the subset of S formed by some prefix of the enumeration.

Randomized Incremental Construction (RIC) attempts to determine all configurations that have no stoppers, i.e. the configurations that are active for S . It does so by enumerating S in random order and keeping track of the active configurations in the corresponding prefix subsets. In the analysis of the performance of RIC algorithms typically the following, or closely related, quantities play a prominent role:

For integer $i \geq 0$ let $A_i = \text{Ex}(X_i)$, where random variable

$$X_i = \sum_{\substack{c \in C \text{ s.t. } c \text{ becomes active} \\ \text{during random enumeration of } S}} (\tau(c) + \sigma(c))^i.$$

Typically A_0 models the expected space requirement and A_1 models the expected running time.

The purpose of this note is to give a short proof of the following:

Theorem 1

$$A_i \leq n^i d^{i+1} \sum_{0 \leq r \leq n} f_0(r) / r^{i+1} \quad \text{for integer } 0 \leq i < \delta$$

with equality if the configuration space is uniform, i.e. $\delta = d$.

Of course $f_0(r)$ is the expected number of configurations active in a random subset of S of size r .

Proof: By definition we have

$$A_i = \sum_{c \in C} \Pr(c \text{ becomes active}) \cdot (\tau(c) + \sigma(c))^i.$$

Since $f_0(r) = \sum_{c \in C} \Pr(c \text{ is active at stage } r)$ the right hand side of the inequality in our theorem can be rewritten as

$$\sum_{c \in C} n^i d^{i+1} \sum_{0 \leq r \leq n} \Pr(c \text{ is active at stage } r) / r^{i+1}.$$

Thus it suffices to consider each $c \in C$ individually. Let $b = \tau(c) \leq d$ and $k = \sigma(c)$ for some fixed c . Since

$$\Pr(c \text{ is active at stage } r) = \Pr(c \text{ becomes active}) \cdot \frac{\binom{r}{b} \binom{n-r}{k}}{\binom{n}{b+k}},$$

our problem reduces to showing that

$$\binom{b+k}{i} = \binom{n}{i} \sum_{0 \leq r \leq n} \frac{\binom{b}{i+1}}{\binom{r}{i+1}} \cdot \frac{\binom{r}{b} \binom{n-r}{k}}{\binom{n}{b+k}}.$$

This replaces falling powers by binomial coefficients, and divides by $i!$ both sides of the equation. In our derivation we will make repeated use of the trinomial identity $\binom{A}{B} \binom{B}{C} = \binom{A}{C} \binom{A-C}{A-B} = \binom{A}{C} \binom{A-C}{B-C}$ and the convolution like identity $\sum_r \binom{r-A-1}{B-A-1} \binom{N-r}{C} = \binom{N-A}{B+C-A}$. With this we get

$$\sum_{0 \leq r \leq n} \frac{\binom{n}{i} \binom{r}{b} \binom{b}{i+1} \binom{n-r}{k}}{\binom{r}{i+1} \binom{n}{b+k}} = \sum_{0 \leq r \leq n} \frac{\binom{n}{i} \binom{r}{i+1} \binom{r-i-1}{b-i-1} \binom{n-r}{k}}{\binom{r}{i+1} \binom{n}{b+k}} = \frac{\binom{n}{i} \binom{n-i}{b+k-i}}{\binom{n}{b+k}} = \frac{\binom{n}{b+k} \binom{b+k}{i}}{\binom{n}{b+k}} = \binom{b+k}{i}.$$

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This note contains a short analysis of certain expected quantities, as they appear in so-called randomized divide-and-conquer. We adopt Mulmuley's terminology of *configuration spaces*, which we briefly review: S is some finite ground set containing n elements; C is some finite set of "configurations" with two functions $st, tr : C \mapsto 2^S$, with $st(c) \cap tr(c) = \emptyset$ for each $c \in C$. The elements in $tr(c)$ are the triggers of configuration c , and the elements in $st(c)$ are its stoppers. We will use $\tau(c) = |tr(c)|$ and $\sigma(c) = |st(c)|$, and we assume that for all c we have $\delta \leq \tau(c) \leq d$ for some constants δ and d . Configuration c is called *active* for $R \subset S$ iff R contains all of c 's triggers but none of its stoppers.

For $1 \leq r \leq n$ let $f_0(r)$ denote the number of configurations active for a random $R \in \binom{S}{r}$. Note that $f_0(r) = \sum_{c \in C} p_r(c)$, where $p_r(c) = \frac{\binom{r}{\tau(c)} \binom{n-r}{\sigma(c)}}{\binom{n}{\tau(c)} \binom{n}{\sigma(c)}}$ is the probability of configuration c being active for a random $R \in \binom{S}{r}$.

For integer $i \geq 0$ and $R \subset S$ let

$$B_i(R) = \sum_{c \text{ active for } R} \sigma(c)^i,$$

and let $B_i(r) = \text{Ex}(B_i(R))$, considering all $R \in \binom{S}{r}$. Note that $B_i(r) = \sum_{c \in C} p_r(c) \sigma(c)^i$.

An important theorem of Clarkson and Shor states that $B_i(r) = O((n-r)/r)^i \phi_0(r)$, where $\phi_0(r) = \max_{1 \leq j \leq r} f_0(j)$. Here we show that if one replaces "max" by "average" in the definition of ϕ_0 the analogous theorem admits a straightforward proof without the detour via f_i 's. In particular we show the following:

Theorem 2

$$B_i(r) \leq \frac{(d+1)^{i+1}}{(r+1)^{i+1}} (n-r)^i \sum_{0 \leq j \leq r} f_0(j)$$

Proof: After substituting the definitions we need to show

$$\sum_{c \in C} p_r(c) \sigma(c)^i \leq \frac{(d+1)^{i+1}}{(r+1)^{i+1}} (n-r)^i \sum_{0 \leq j \leq r} \sum_{c \in C} p_j(c).$$

Rearranging the sums on the right hand side shows that it suffices to consider each $c \in C$ individually. So let c be some configuration with $b = \tau(c) \leq d$ and $k = \sigma(c)$. It suffices to prove

$$p_r(c) k^i \leq \frac{(b+1)^{i+1}}{(r+1)^{i+1}} (n-r)^i \sum_{0 \leq j \leq r} p_j(c).$$

Dividing both sides by $i!$, replacing all rising and falling powers by binomial coefficients, moving them to the left hand side, and substituting values for the p_j 's while omitting their common denominator reduces this inequality to

$$\frac{\binom{r+i+1}{i+1} \binom{r}{b} \binom{n-r}{k} \binom{k}{i}}{\binom{b+i+1}{i+1} \binom{n-r}{i}} \leq \sum_{0 \leq j \leq r} \binom{j}{b} \binom{n-j}{k}.$$

Applying the trinomial identity $\binom{A}{B} \binom{B}{C} = \binom{A}{C} \binom{A-C}{B-C}$ twice, the left hand side reduces to

$$\frac{\binom{r+i+1}{b+i+1} \binom{b+i+1}{i+1} \binom{n-r}{i} \binom{n-r-i}{k-i}}{\binom{b+i+1}{i+1} \binom{n-r}{i}} = \binom{r+i+1}{b+i+1} \binom{n-r-i}{k-i}.$$

Thus we are left with proving that

$$\binom{r+i+1}{b+i+1} \binom{n-r-i}{k-i} \leq \sum_{0 \leq j \leq r} \binom{j}{b} \binom{n-j}{k}.$$

The validity of this inequality can be seen by noting that the left hand side counts the set T of all binary strings of length $n+1$ with exactly $b+k+1$ digits '1' of which exactly $b+i+1$ are in the first $r+i+1$ positions. Each string in T has the property that its $(b+1)$ -st '1' appears on position $j+1$ with $j \leq r$. But the right hand side counts exactly all binary strings of length $n+1$ with exactly $b+k+1$ ones that have that property. ■